Age of Gray Matters: Neuroprediction of Recidivism

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Abstract

Age is one of the best predictors of antisocial behavior. Risk models of recidivism often combine chronological age with demographic, social and psychological features to aid in judicial decision-making. Here we use independent component analyses (ICA) and machine learning techniques to demonstrate the utility of using brain-based measures of cerebral aging to predict recidivism. First, we developed a brain-age model that predicts chronological age based on structural MRI data (n=1332). We then test the model's ability to predict recidivism in a new sample of offenders with outcome data (n=93). Consistent with hypotheses, brain-age measures of the inferior frontal cortex and anterior-medial temporal lobes (i.e., amygdala) outperformed chronological age in prediction of recidivism. These results suggest that brain measures may complement or even supplant traditional proxy measures when they more precisely assess constructs of interest and improve prediction of future behavior. This work also identifies the brain systems that contribute to recidivism which has implications for treatment development.

Keywords: Neuroprediction, age, recidivism, antisocial, MRI

A practical approach for differentiating risk levels among offenders is to develop algorithms that identify variables that predict how likely inmates are to commit another crime after their release from prison. Meta-analyses have identified several key risk variables including criminogenic needs, demographics, social achievement, socio-economic status, and intelligence (Gendreau, Little, & Goggin, 1996). Additional research has identified empirically derived static (e.g., past criminal history, offense type) and dynamic (e.g., impulsivity, drug use, social support) risk factors that have led to significant improvements in predicting future antisocial behavior (Douglas, Webster, Hard, Eaves, & Ogloff, 2002; Harris, Rice, & Quinsey, 1993; Yang, Wong, & Coid, 2010). These risk assessment procedures are useful in judicial decision-making and in creating release plans that minimize risk factors (e.g., substance abuse) and accentuate protective factors (e.g., social support, stable employment).

Developments in bio-psycho-social models have identified risk variables with strong relevance to antisocial behavior. Age for example, is a powerful variable in the prediction the likelihood for antisocial behavior (Gendreau et al., 1996). Indeed, if we consider the release of two inmates from prison, a 25-year-old and a 35-year-old, all else being equal, the 25-year-old is roughly 25% more likely to be re-incarcerated within five years following their release than is the 35-year old (Durose, Cooper, & Snyder, 2014). Age also features prominently in societal decisions about holding people accountable for their behavior, as our treatment of juvenile offenders is categorically different than that of adults.

Chronological age may, however, may be an imprecise measure in these risk equations. That is, within the spectrum of all 25-year olds, some of the cohort may be lower than average risk and some may be higher than average risk to reoffend. Thus, chronological age does not account for individual differences in aging processes.

Recent developments have shown that biological aging of the brain can be quantified using MRI techniques (Brown et al., 2012). Here we develop a brain-age model using multivariate analyses of structural MRI data and apply our brain-age measures to a new cohort of male offenders. We test whether our brain-age measures predict antisocial outcomes better than chronological age. To our knowledge, this is the first attempt to develop a brain maturation model to distinguish individuals who are more or less likely to reoffend following release from prison.

We present two analyses. First, independent component analysis (ICA) was applied to structural brain imaging data (often called source-based morphometry or SBM; (Calhoun, 2001; Caprihan et al., 2011; Xu, Groth, Pearlson, Schretlen, & Calhoun, 2009)). ICA components were estimated from a large sample (n = 1332) of incarcerated males ranging in age from 12-65 years. We found 19 brain volume and 19 density components (Figure 1a/b; Table 1 & 2) accounting for 68.2% and 71.0%, respectively, of the variance in chronological age.

In a separate experiment, we extracted brain-age-related ICA components from an independent sample (n = 93; (Aharoni et al., 2013; Steele et al., 2015)) and used machine-learning techniques to identify those age-related components that predicted recidivism. Two volume[1] and one density[2] ICs were identified to be useful in predicting rearrest (see Figure 2). Cox proportional hazards regression models predicting rearrest using the ICA derived brain-age components (along with other actuarial risk variables) were computed.

Method

[1] Volume components identified with feature selection: Components 14 and 24

[2] Density components identified with feature selection: Components 4, 16, 21, 24, and 26.

All study procedures described below were carried out in accordance with The Code of Ethics of the World Medical Association (Declaration of Helsinki) for experiments involving humans, and approved by the University of New Mexico Institutional Review Board. Individuals 18 years of age or older provided written informed consent and individuals younger than 18 years of age provided written informed assent in conjunction with parent/guardian written informed consent.

Participants

Participants included 1332 male offenders (sample 1) ranging from 12 to 65 years of age (*M* = 30.5, *SD* = 11.46) and 93 male offenders (sample 2; 20 to 52 years of age (*M* = 32.94, *SD* = 7.83)) for which follow-up recidivism data and high-resolution MRI scan data was available (Aharoni et al., 2013). None of participants in Experiment 2 were included in Experiment 1. Based on the NIH racial and ethnic classification, 4% of the sample self-identified as American Indian/Alaskan Native, 12% as Black/African American, 1% as Native Hawaiian or other Pacific Islander, 29% as White, 22% as Hispanic/Latino, 1% identified as more than one race, and 30% chose not to respond.

Full-scale IQ was estimated using the Vocabulary and Matrix Reasoning sub-tests of the Wechsler Intelligence Scale for Children – 4th Edition (WISC-IV; (Wechsler, 2003)) for participants younger than 18 years of age, and the Wechsler Adult Intelligence Scale – 3rd Edition (WAIS-III; (Wechsler, 1997)) for participants older than 18 years of age (*M* = 97.13, *SD* = 13.97)). Mental illness and substance use was assessed using the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS; (Kaufman, Birmaher, & Brent, 1997)) for participants younger than 18 years of age and the Structured Clinical Interview for DSM-IV Axis I Disorders – Patient Version (SCID I-P; (First, Spitzer, Gibbon, & Williams, 1997)).

Participants were excluded from analyses if they had a full-scale IQ less than 70 or were unable to complete a 'research consent test' examining their understanding of the research study, reported a TBI accompanied with a significant loss of consciousness, or history of personal or familial bipolar or psychotic disorders.

MRI Acquisition

T1-weighted MRI scans were acquired on the Mind Research Network (MRN) Siemens 1.5T Avanto mobile scanner stationed at the prisons using a multi-echo MPRAGE pulse sequence (repetition time = 2530 ms, echo times = 1.64 ms, 3.50 ms, 5.36 ms, 7.22 ms, inversion time = 1100 ms, flip angle = 7°, slice thickness = 1.3 mm, matrix size = 256 × 256) yielding 128 sagittal slices with a resolution of 1.0 mm × 1.0 mm x 1.0mm. Images were spatially normalized to the Montreal Neurological Institute (MNI) template using SPM12, segmented into gray matter, white matter, and cerebrospinal fluid. Both grey matter volume and density were extracted for analyses. A Jacobian modulation was performed to preserve total volume (Ashburner & Friston, 2000, 2005). Grey matter images were resampled to 2 × 2 × 2 mm and smoothed with a 10 mm full-width at half-maximum (FWHM) Gaussian kernel.

Analysis Experiment 1

ICA of structural MRI (i.e., SBM) was computed from sample 1 (Calhoun, 2001; Caprihan et al., 2011; Xu et al., 2009). Thirty volume and density components were extracted using the GIFT toolbox (http://mialab.mrn.org/software/gift). Loading coefficients were extracted for each IC and for each participant. These coefficients were then used in a stepwise linear regression as independent variables (IVs) with chronological age as the dependent variable (DV).

Experiment 2

Experiment 2 is designed to use the ICs from Experiment 1 that account for chronological age to predict future rearrest. Machine learning techniques were used to identify brain ICs that predicted recidivism from both brain volume and density ICs. Cox proportional hazards regression models using chronological and neural age measures were computed and compared.

Participants

The sample 2 participants were followed after release from prison and tracked from 2007 to 2010. The average follow-up period was 21.96 months (range: 1.51 to 49.55 months; see (Aharoni et al., 2013; Steele et al., 2015). Other variables collected included psychopathy, which was assessed using the Hare Psychopathy Checklist-Revised (PCL-R; (Hare, 2003)(*M* Total Score = 23.41, *SD* = 6.89; scores for n=11 unavailable). Full-scale IQ was estimated using the Vocabulary and Matrix Reasoning sub-tests of the WAIS-III (*M* = 94.27, *SD* = 12.75; n=11 subjects did not complete IQ). Mental illness was assessed using the SCID I-P. An average drug abuse/dependence measure was calculated from the following drug classes: sedatives (6% met for dependence), cannabis (61% met for dependence), stimulants (49% met for dependence), opioids (23% met for dependence), cocaine (56% met for dependence), and hallucinogens (9% met for dependence); substance use scores were unavailable for *n* = 13 participants. Additionally, 54% met criteria for alcohol dependence. Based on the NIH racial and ethnic classification, 5% of the sample self-identified as American Indian/Alaskan Native, 6% as Black/African American, 22% as White, 41% as Hispanic/Latino, 1% identified as more than one race, and 25% chose not to respond.

Analysis Experiment 2

Using the neural age measures identified in Experiment 1, loading parameters for the ICs were estimated from participants in Experiment 2 using spatio-temporal (dual) regression (STR; (Erhardt, 2011)). In this way, the data in Experiment 2 was projected on to the ICA calculated on participants from Experiment 1. This step avoids bias in IC definition as none of the participants in Experiment 2 were used to define the ICs related to age.

The 19 brain volume measures derived in Experiment 1 account 68.2% of the variance in age in Experiment 1 and 59.0% of variance in age in Experiment 2. The 19 density measures derived in Experiment 1 account for 67.1 percent of variance in age in Experiment 2 compared to 71.0% in Experiment 1. When combined into a single regression the 19 volume and 19 density variables account for 75.0 percent of variance in age in Experiment 2.

In an attempt to reduce model complexity in predicting recidivism, we used a sequential K best feature selection with two nested cross validation loops (10-fold for outer loop and 3-fold for inner loop) using Support Vector Machines (SVM) as a criterion function to identify which components were most useful in predicting rearrest (Jain, Duin, & Mao, 2000). The 10-fold cross validation outer loop cycles across subjects and the 3-fold inner loop cycles across a set of parameters for the SVM model. The goal of feature selection is to prune the set of IC (or behavioral) variables to identify the best discriminative set. Non-informative features add noise and dimensionality to the data that result in poor classification performance. Sequential K best feature selection selects a set of variables from the input feature set for classifying individuals into groups (i.e., rearrested [n = 50] versus non-rearrested [n = 43]). This feature selection is a method that iterates selecting K best features based on ANOVA F-scores between the two classes of interest. After this step, the scores are averaged across the 10 folds and the maximum resulting accuracy score is selected as the best K features. Support Vector Machine classifier is a binary classifier that aims at finding a hyperplane that maximizes the margin between the two classes. This was implemented using built in feature selection functions in Python using the Scikit-Learn library. Using this method, two volume[3] and five density[4] ICs were identified to be useful in predicting rearrest. These neural age measures identified in this feature selection step were used in Cox proportional hazards regression predicting time to rearrest.

Cox regression takes 'time at risk' into account by using time to rearrest as the outcome variable, calculated as the number of days between release from incarceration and the rearrest date, or the follow-up date for those who were not rearrested. Arrest data was collected using a nationwide commercial search company (SSC, Inc). Those who were not rearrested are included in the analyses and considered to be 'censored' cases, meaning they potentially could still reoffend, accounting for variable lengths of follow-up. Reliability of the Cox regressions was assessed using bootstrapping with 9,999 iterations. Cox regression models were computed with covariates from (Aharoni et al., 2013): PCL-R Factor 1, PCL-R Factor 2, the interaction of PCL-R Factor 1 and PCL-R Factor 2, drug and alcohol dependence, false alarm rate from the Go/No Go task, ACC activity extracted from the Go/No Go task, and chronological age, as well the brain age measures identified here. Independent samples *t*-tests were performed on these variables to assess differences between offenders who were rearrested and those who were not.

A total of eight Cox proportional hazards regressions were computed with rearrest as the binary DV and time-to-rearrest as the continuous DV. The models were:

Model 1) chronological age (replicating the model in (Aharoni et al., 2013) minus three participants who were excluded due to motion contaminated MRIs);

[3] Volume components identified with feature selection: Components 14 and 24

[4] Density components identified with feature selection: Components 4, 16, 21, 24, and 26.

Model 2) chronological age and covariates from (Aharoni et al., 2013) (PCL-R Factor 1, PCL-R Factor 2, the interaction of PCL-R Factor 1 and PCL-R Factor 2, drug and alcohol dependence, false alarm rate from the Go/No Go task, Go/No Go ACC activity); this model replicated that in (Aharoni et al., 2013) minus three participants who were excluded due to motion contaminated MRIs).

Model 3) brain volume measures of age-related components selected to be useful in predicting rearrest (i.e., 2 volume components);

Model 4) the two brain volume variables in Model 3 and (Aharoni et al., 2013) covariates (PCL-R Factor 1, PCL-R Factor 2, the interaction of PCL-R Factor 1 and PCL-R Factor 2, drug and alcohol dependence, false alarm rate from the Go/No Go task, ACC activity extracted from the Go/No Go task);

Model 5) brain and psychological variables in Model 4 and chronological age;

Model 6) brain age density variables (5 density components)

Model 7) brain density components related to age and covariates from (Aharoni et al., 2013) (PCL-R Factor 1, PCL-R Factor 2, the interaction of PCL-R Factor 1 and PCL-R Factor 2, drug and alcohol dependence, false alarm rate from the Go/No Go task, Go/No Go ACC activity);

Model 8) variables in Model 7 and chronological age.

In combination, these models allow for a full picture of how well brain age measures predict rearrest, whether brain age adds incrementally to models with chronological age, and if chronological age is necessary in models with brain age measures. Goodness-of-fit were calculated for each model and compared between models by calculating the Akaike Information Criterion (AIC; (Akaike, 1992)), Likelihood ratio test and Score log rank test.

Results

Experiment 1

Stepwise linear regressions were calculated for brain volume measures predicting chronological age. Nineteen brain volume and 19 density ICs (Figure 1a/b; Table 1 & 2) accounting for 68.2% and 71.0%, respectively, of the variance in chronological age. A four-dimensional NIFTI file containing these ICs is provided in the supplemental material.

Figure 1a. Source-based morphometry (SBM) components of volume identified to account for 68.2% of the variance of chronological age. These 19 components define neural age. Components 14 (temporal pole) and 24 (inferior temporal gyrus) were feature selected to be beneficial in predicting rearrest. Components 3 (cerebellum), 4 (inferior and superior parietal gyrus and occipital lobe), 5 (putamen), 6 (cerebellum), 8 (superior and middle frontal gyrus and supplementary motor area), 10 (precentral and postcentral gyrus), 11 (superior parietal gyrus and occipital lobe), 12 (inferior parietal and postcentral gyrus), 14 (temporal pole), 15 (middle temporal gyrus), 16 (cerebellum and lingual gyrus), 19 (fusiform and inferior temporal gyrus), 20 (orbitofrontal cortex and insula), 22 (cerebellum, hippocampus, and amygdala), 25 (middle and inferior temporal gyrus), 26 (cerebellum), 27 (inferior and middle frontal gyrus), and 28 (precentral gyrus) were not selected to be beneficial in predicting rearrest.

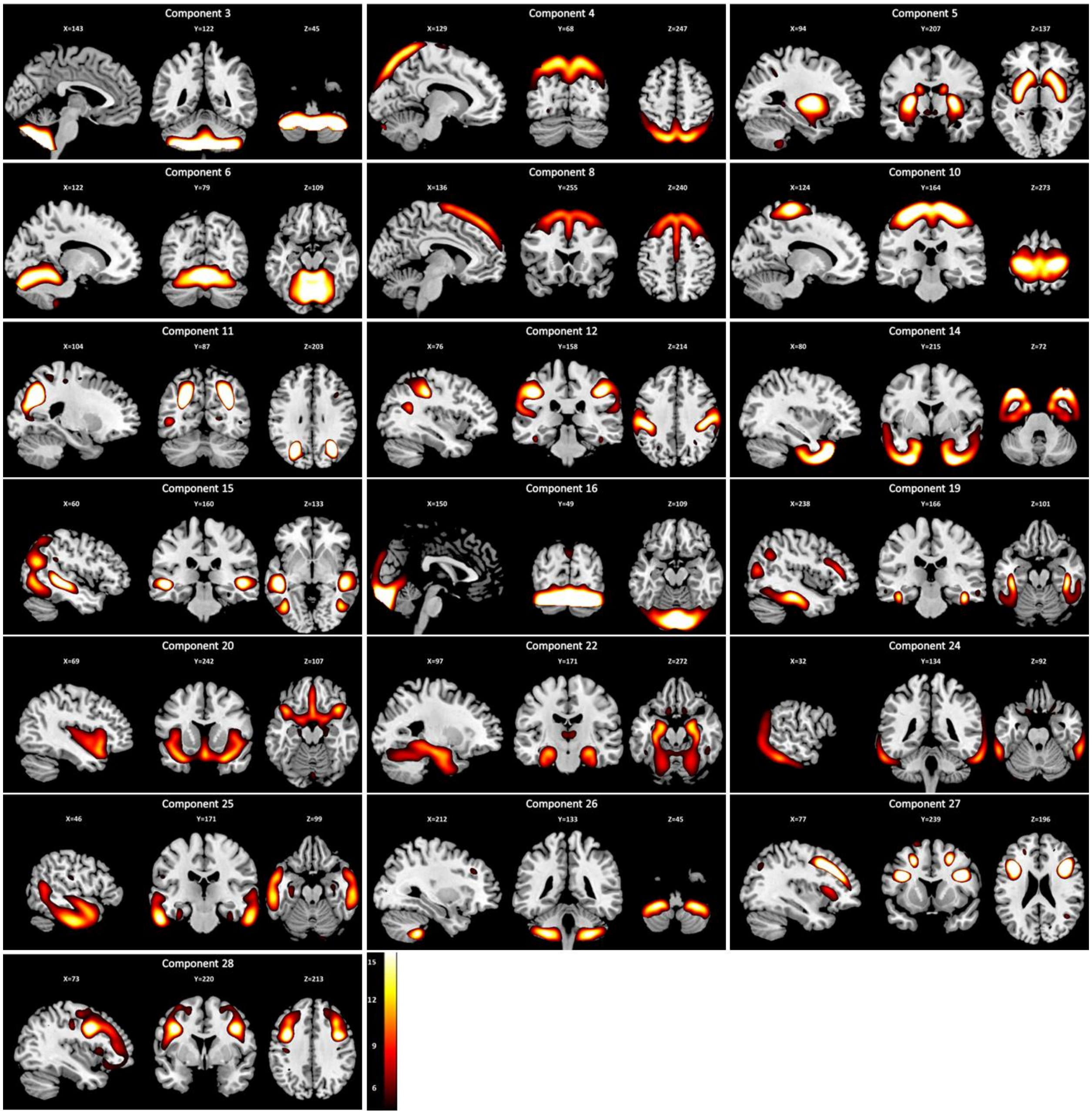

Component 3
X=143
Y=122
Z=45
Component 4
X=129
Y=68
Z=247
Component 5
X=94
Y=207
Z=137
Component 6
X=122
Y=79
Z=109
Component 8
X=136
Y=255
Z=240
Component 10
X=124
Y=164
Z=273
Component 11
X=104
Y=87
Z=203
Component 12
X=76
Y=158
Z=214
Component 14
X=80
Y=215
Z=72
Component 15
X=60
Y=160
Z=133
Component 16
X=150
Y=49
Z=109
Component 19
X=238
Y=166
Z=101
Component 20
X=69
Y=242
Z=107
Component 22
X=97
Y=171
Z=272
Component 24
X=32
Y=134
Z=92
Component 25
X=46
Y=171
Z=99
Component 26
X=212
Y=133
Z=45
Component 27
X=77
Y=239
Z=196
Component 28
X=73
Y=220
Z=213
15
12
9
6

Figure 1b. Source-based morphometry (SBM) components of density identified to account for 71.0% of the variance of chronological age. These 19 components define neural age. Components 4 (angular gyrus), 16 (inferior parietal gyrus), 21 (temporal pole), 24 (cerebellum), and 26 (occipital lobe) were feature selected to be beneficial in predicting rearrest. Components 2 (precentral and postcentral gyrus), 3 (middle frontal gyrus), 10 (cerebellum), 12 (cerebellum), 14 (cerebellum, hippocampus, and amygdala), 17 (cerebellum), 18 (putamen), 19 (superior parietal gyrus and precuneus), 20 (fusiform gyrus, calcarine fissure, lingual gyrus, and occipital lobe), 25 (precuneus and calcarine fissure), 27 (fusiform gyrus, calcarine fissure, lingual gyrus, and occipital lobe), 28 (superior and middle frontal gyrus), 29 (middle and inferior temporal gyrus), and 30 (orbitofrontal cortex) were not selected to be beneficial in predicting rearrest.

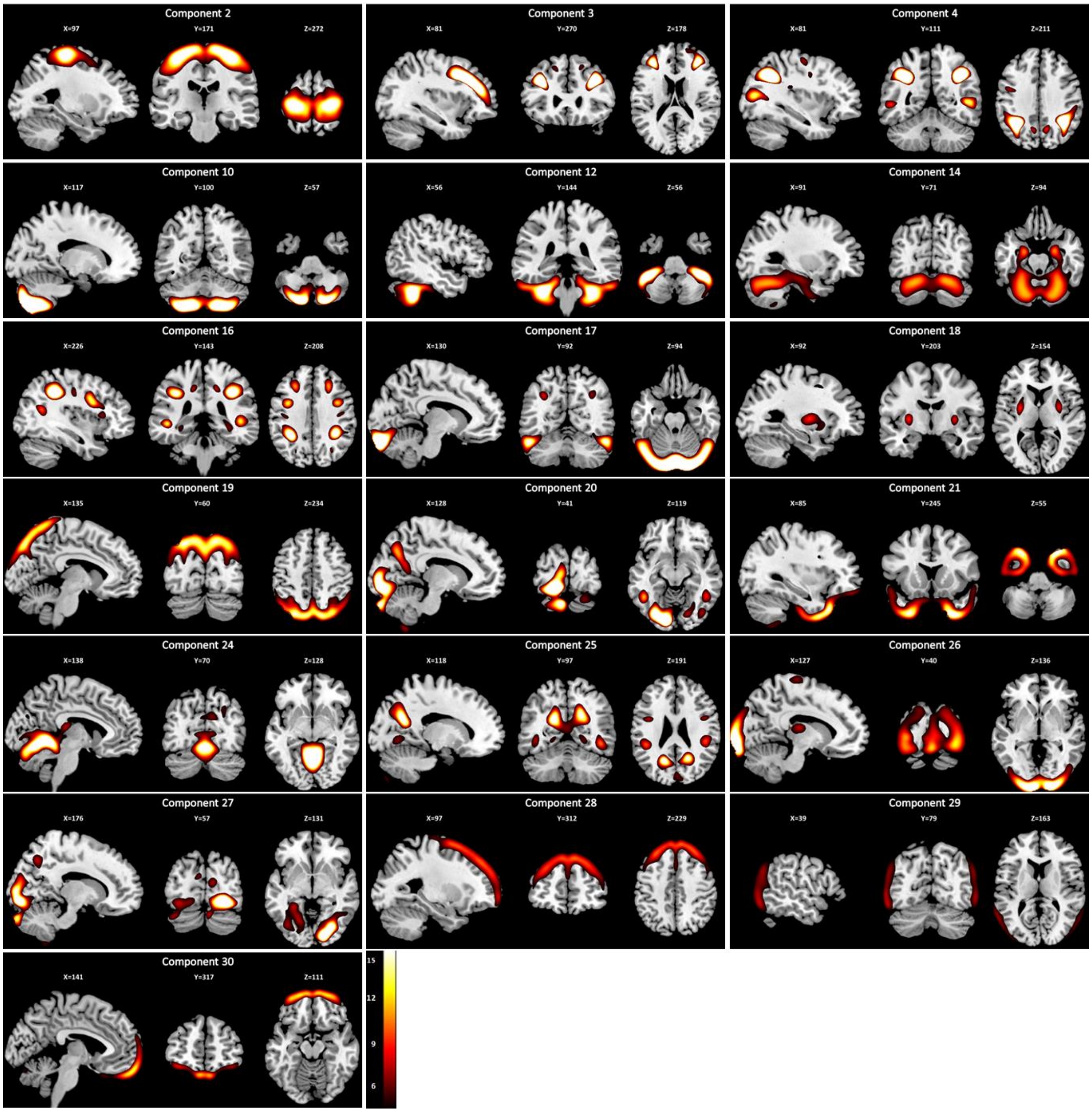

Component 2
X=97
Y=171
Z=272
Component 3
X=81
Y=270
Z=178
Component 4
X=81
Y=111
Z=211
Component 10
X=117
Y=100
Z=57
Component 12
X=56
Y=144
Z=56
Component 14
X=91
Y=71
Z=94
Component 16
X=226
Y=143
Z=208
Component 17
X=130
Y=92
Z=94
Component 18
X=92
Y=203
Z=154
Component 19
X=135
Y=60
Z=234
Component 20
X=128
Y=41
Z=119
Component 21
X=85
Y=245
Z=55
Component 24
X=138
Y=70
Z=128
Component 25
X=118
Y=97
Z=191
Component 26
X=127
Y=40
Z=136
Component 27
X=176
Y=57
Z=131
Component 28
X=97
Y=312
Z=229
Component 29
X=39
Y=79
Z=163
Component 30
X=141
Y=317
Z=111
15
12
9
6

Table 1: Linear stepwise regressions predicting chronological age with volume and density SBM components

Volume				Density			
Predictors	β	t	Sig.	Predictors	B	t	Sig.
Significant Predictors				**Significant Predictors**			
10	-0.186	-6.887	< .001	2	-0.200	- 6.769	< .001
16	-0.308	-12.850	< .001	18	-0.440	-16.047	< .001
22	0.431	20.151	< .001	29	-0.291	-10.134	< .001
15	-0.167	-6.078	< .001	26	0.114	6.116	< .001
28	-0.147	-5.310	< .001	24	-0.197	-11.104	< .001
5	-0.160	-7.911	< .001	12	-0.203	-8.725	< .001
24	-0.293	-10.261	< .001	10	0.113	7.057	< .001
4	0.231	7.776	< .001	30	-0.086	-4.568	< .001
6	-0.153	-6.927	< .001	3	-0.067	-3.541	< .001
19	0.115	5.388	< .001	19	0.158	4.577	< .001
11	-0.067	-3.693	< .001	20	0.076	4.574	< .001
27	0.061	3.785	< .001	27	0.060	3.882	< .001
20	0.145	5.215	< .001	16	-0.089	-3.735	< .001
12	-0.099	-3.560	< .001	25	0.060	3.542	< .001
3	0.073	3.378	< .001	28	-0.098	-3.172	.002
14	0.101	4.802	.001	21	0.102	3.653	< .001
26	0.083	3.500	< .001	17	-0.070	-3.132	.002
8	-0.130	-4.129	< .001	14	0.057	2.865	.004
25	-0.070	-2.858	.004	4	0.041	2.607	.009
Nonsignificant Predictors				**Nonsignificant Predictors**			
1	0.012	0.505	.614	1	0.025	0.609	.542
2	0.009	0.472	.637	9	-155.645	-1.571	.116
7	-0.037	-1.653	.098	11	-0.035	-1.445	.149
9	0.009	0.402	.688	13	0.026	1.664	.096
13	0.009	0.554	.580	15	-0.015	-0.858	.391
17	-0.026	-0.951	.342	22	0.010	0.599	.549
18	0.002	0.099	.921	23	0.003	0.204	.838
23	0.042	1.682	.093				
29	0.024	1.384	.167				
30	0.037	1.920	.055				

Note. On the left side of the table is the final step of a linear stepwise regression predicting chronological age with volume SBM components. Significant components are listed in the order in which they were selected for the model. These 19 components account for 68.2% of the variance in chronological age and, taken together, are a neural measure of age. $R^2 = .682$, $R = .826$, $p < .001$. Component numbers are listed for SBM components. On the right side of the table is the final step of a linear stepwise regression predicting chronological age with density SBM components. Significant components are listed in the order in which they were selected for the model. These 19 components account for 71.0% of the variance in chronological age and, taken together, are a neural measure of age. $R^2 = .710$, $R = .843$ ($p < .001$). Components 7 and 8 are not included in above table due to having tolerance values of .000. Tolerance is an indication of the percent of variance in the predictor that cannot be accounted for by the other predictors; hence, very small values (e.g., .000) indicate a predictor is redundant. Component numbers are listed for SBM components.

Experiment 2

The first two models confirmed previous analyses (Aharoni et al., 2013). Prediction Model 1, which only included chronological age, was marginally significant ($p = .088$; Table 2). Model 2 was significant ($p = .030$) and PCL-R Factor 2 ($p = .010$), ACC activity ($p = .001$), and chronological age ($p = .048$) were unique predictors (Table 2). Chronological age did not differ between those who re-offended and those who did not, $t(91) = 1.60$, $p = .113$. ACC activity did differ between groups ($t(91) = 2.01$, $p = .047$) and PCL-R Factor 2 was marginally different between groups ($t(85) = 1.82$, $p = .073$).

Models 3-5, which used brain age estimated using volume measures, were significant. Model 3 ($p = .003$), which included only brain volume components, found that Component 14 (temporal pole; $p = .078$), and 24 (interior temporal gyrus; $p = .074$) (Table 2b) were marginally significant predictors. Model 4 ($p = .005$) included PCL-R Factor 2 ($p = .023$), and ACC activity ($p = .022$). Component 14 (temporal pole; $p = .079$) was a marginal predictor (Table 2b). Model 5 ($p = .001$) included PCL-R Factor 2 ($p = .028$), ACC activity ($p = .004$), chronological age ($p = .004$) and Component 24 (inferior temporal gyrus; $p = .029$) as unique predictors (Table 2b). Component 14 (temporal pole; $p = .089$) was a marginal predictor.

Using t-tests, less gray matter volume was identified for the rearrested group, compared to the not-rearrested group, in Components 14 (temporal pole) and 24 (inferior temporal gyrus), $t(91) = 2.74$, $p = .007$, $t(91) = 3.14$, $p = .002$, respectively.

Models 6-8, which used brain age estimated using density measures, were significant. Model 6 ($p = .004$) included component 21 (inferior frontal/temporal pole; $p = .05$) as a predictor (Table 2c). Model 7 ($p = .003$) included PCL-R Factor 2 ($p = .012$), ACC activity ($p = .002$), and Component 21 (inferior frontal/temporal pole; $p = .020$), as unique predictors (Table 2c). Model 8 ($p = .002$) included PCL-R Factor 2 ($p = .008$), ACC activity ($p = .001$), chronological age ($p = .035$), and Component 21 (inferior frontal/temporal pole; $p = .005$) as unique predictors (Table 2c).

Table 2a: Preliminary Cox proportional hazards regressions

Predictor	*B*	*Boot-strapped B*	*SE (B)*	*Boot-strapped SE (B)*	*p*-value	exp[B]	CI (95%) for exp[B]	*Boot-strapped CI (95%) for exp[B]*
Model 1								
Rel. Age	-0.03	-0.03	-0.02	-0.0002	.088	0.97	0.93 – 1.01	0.93 – 1.00
Model 2								
PCL-R F1	-0.06	-0.06	0.07	6.73×10^{-3}	.396	0.94	0.82 – 1.08	0.80 – 1.07
PCL-R F2	**-2.60**	**-2.60**	**1.01**	$\mathbf{-2.90 \times 10^{-1}}$	**.010**	**0.07**	**0.01 – 0.53**	**0.01 – 0.40**
PCL-R Int.	0.07	0.07	0.25	5.78×10^{-3}	.788	1.07	0.66 – 1.73	0.65 – 1.88
Drug	-0.19	-0.19	0.37	-1.25×10^{-2}	.609	0.83	0.40 – 1.72	0.34 – 1.92
Alcohol	0.11	0.11	0.20	1.36×10^{-2}	.591	1.11	0.75 – 1.66	0.73 – 1.80
FA	-0.01	-0.01	0.01	-1.66×10^{-5}	.571	0.99	0.97 – 1.02	0.97 – 1.02
ACC	**-0.69**	**0.69**	**0.21**	$\mathbf{5.94 \times 10^{-2}}$	**.001**	**0.50**	**0.34 – 0.75**	**0.29 – 0.72**
Rel. Age	**-0.04**	**-0.04**	**0.96**	$\mathbf{-4.48 \times 10^{-3}}$	**.048**	**0.96**	**0.92 – 1.00**	**0.91 – 1.00**

Note. Results of Cox proportional hazards regression analyses examining the predictive effect chronological age (model 1) and chronological age with covariates (model 2) on rearrest are presented. Model 1: Wald(1) = 2.92, p = .088; Likelihood Ratio(1) = 3.02, p = 0.082; Score(logrank)(1) = 2.95, p = 0.086. Model 2: Wald(8) = 17.04, p = .030; Likelihood Ratio(8) = 19.19, p = 0.014; Score(logrank)(8) = 17.87, p = 0.022. Bold variables are unique predictors within the model. Rel. Age is the participant's age when released from the correctional facility; PCL-R F1 and F2 refer to Factor 1 and Factor 2 scores from the Psychopathy Checklist – Revised (PCL-R) (Hare, 2003); PCL-R Int. refers to the PCL-R interaction term, formed by multiplying PCL-R Factor 1 and Factor 2 scores together; Drug refers to the participant's average use of the following drug classes: sedatives, cannabis, stimulants, opioids, cocaine, and hallucinogens collected from the Scheduled Clinical Interview for DSM-IV Axis I Disorders – Patient Version (SCID I/P) (First, Gibbon, Spitzer, Williams, & Benjamin, 1997) and the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS) (Kaufman et al., 1997); Alcohol refers to the participant's average use of alcohol collected from the SCID I/P and K-SADS; FA refers to the false alarm rate to NoGo stimuli; ACC refers to dorsal anterior cingulate cortex mean activation.

Table 2b: Cox proportional hazards regressions with volume SBM components

Predictor	*B*	*Boot-strapped B*	*SE (B)*	*Boot-strapped SE (B)*	*p*-value	exp[B]	CI (95%) for exp[B]	*Boot-strapped CI (95%) for exp[B]*
Model 3								
04	3.45×10^{-1}	3.84×10^{-1}	3.00×10^{-1}	3.38×10^{-1}	.249	1.41×10^{0}	7.85×10^{-1} – 2.54×10^{0}	-3.56×10^{-1} – 9.69×10^{-1}

06	3.38×10^{-1}	3.76×10^{-1}	2.34×10^{-1}	2.67×10^{-1}	.148	1.40×10^{0}	8.87×10^{-1} – 2.22×10^{0}	-2.25×10^{-1} – 8.22×10^{-1}
10	-2.37×10^{-1}	-2.72×10^{-1}	2.32×10^{-1}	2.64×10^{-1}	.307	7.89×10^{-1}	5.01×10^{-1} – 1.24×10^{0}	-7.19×10^{-1} – 3.17×10^{-1}
14	-3.87×10^{-1}	-4.36×10^{-1}	2.21×10^{-1}	2.62×10^{-1}	.080	6.79×10^{-1}	4.40×10^{-1} – 1.05×10^{0}	-8.53×10^{-1} – 1.76×10^{-1}
16	1.47×10^{-1}	1.65×10^{-1}	2.26×10^{-1}	2.61×10^{-1}	.516	1.16×10^{0}	7.44×10^{-1} – 1.80×10^{0}	-3.83×10^{-1} – 6.40×10^{-1}
19	-1.15×10^{-1}	-1.25×10^{-1}	2.30×10^{-1}	2.46×10^{-1}	.616	8.91×10^{-1}	5.68×10^{-1} – 1.40×10^{0}	-5.89×10^{-1} – 3.38×10^{-1}
20	1.97×10^{-1}	2.25×10^{-1}	3.01×10^{-1}	3.38×10^{-1}	.512	1.22×10^{0}	6.76×10^{-1} – 2.20×10^{0}	-4.94×10^{-1} – 8.32×10^{-1}
22	-4.75×10^{-1}	-5.35×10^{-1}	2.66×10^{-1}	2.93×10^{-1}	.074	6.22×10^{-1}	3.69×10^{-1} – 1.05×10^{0}	-9.92×10^{-1} – 1.60×10^{-1}
24	-5.71×10^{-1}	-6.42×10^{-1}	3.11×10^{-1}	3.69×10^{-1}	.066	5.65×10^{-1}	3.08×10^{-1} – 1.04×10^{0}	-1.22×10^{-1} – 2.24×10^{-1}
25	-1.60×10^{-1}	-1.82×10^{-1}	2.64×10^{-1}	3.10×10^{-1}	.545	8.52×10^{-1}	5.08×10^{-1} – 1.43×10^{0}	-7.44×10^{-1} – 4.70×10^{-1}
26	-3.74×10^{-1}	-4.18×10^{-1}	2.70×10^{-1}	2.92×10^{-1}	.166	6.88×10^{-1}	4.05×10^{-1} – 1.17×10^{0}	-9.02×10^{-1} – 2.42×10^{-1}
Model 4 PCL-R F1	-4.22×10^{-2}	-4.73×10^{-2}	6.65×10^{-2}	-4.79×10^{-3}	.526	9.59×10^{-1}	8.42×10^{-1} – 1.09×10^{0}	8.24×10^{-1} – 1.09×10^{0}
PCL-R F2	**-2.32×10^{0}**	**-2.61×10^{0}**	**1.02×10^{0}**	**-2.88×10^{-1}**	**.023**	**9.87×10^{-2}**	**1.35×10^{-2} – 7.25×10^{-1}**	**8.21×10^{-3} – 5.32×10^{-1}**
PCL-R Int.	7.40×10^{-2}	7.86×10^{-2}	2.34×10^{-1}	7.02×10^{-3}	.752	1.08×10^{0}	6.81×10^{-1} – 1.70×10^{0}	6.52×10^{-1} – 1.84×10^{0}
Drug	4.48×10^{-2}	6.02×10^{-2}	3.83×10^{-1}	1.05×10^{-2}	.907	1.05×10^{0}	4.94×10^{-1} – 2.21×10^{0}	4.13×10^{-1} – 2.80×10^{0}
Alcohol	1.62×10^{-3}	1.91×10^{-4}	1.96×10^{-1}	-2.97×10^{-3}	.993	1.00×10^{0}	6.83×10^{-1} – 1.47×10^{0}	6.32×10^{-1} – 1.59×10^{0}
FA	-5.37×10^{-3}	-5.53×10^{-3}	1.24×10^{-2}	1.39×10^{-4}	.664	9.95×10^{-1}	9.71×10^{-1} – 1.02×10^{0}	9.68×10^{-1} – 1.02×10^{0}
ACC	**-4.36×10^{-1}**	**-4.78×10^{-1}**	**1.90×10^{-1}**	**-4.25×10^{-2}**	**.022**	**6.47×10^{-1}**	**4.45×10^{-1} – 9.39×10^{-1}**	**3.87×10^{-1} – 9.49×10^{-1}**

14	-3.4×10^{-1}	-3.73×10^{-1}	1.93×10^{-1}	2.21×10^{-1}	.079	7.12×10^{-1}	4.87×10^{-1} – 1.04×10^{0}	-7.40×10^{-1} – 1.29×10^{-1}
24	-2.68×10^{-1}	-3.03×10^{-1}	1.97×10^{-1}	2.31×10^{-1}	.173	7.64×10^{-1}	5.20×10^{-1} – 1.12×10^{0}	-6.89×10^{-1} – 2.16×10^{-1}
Model 5 PCL-R F1	-3.39×10^{-2}	-4.01×10^{-1}	7.42×10^{-2}	-5.84×10^{-3}	.647	9.67×10^{-1}	8.36×10^{-1} – 1.19×10^{0}	8.19×10^{-1} – 1.11×10^{0}
PCL-R F2	**-2.32×10^{0}**	**-2.68×10^{0}**	**1.05×10^{0}**	**-3.43×10^{-1}**	**.028**	**9.87×10^{-2}**	**1.25×10^{-2} – 7.77×10^{-1}**	**6.17×10^{-2} – 5.75×10^{-1}**
PCL-R Int.	2.49×10^{-1}	2.77×10^{-1}	2.53×10^{-1}	2.79×10^{-2}	.325	1.28×10^{0}	7.81×10^{-1} – 2.11×10^{0}	7.57×10^{-1} – 2.39×10^{0}
Drug	2.50×10^{-1}	2.90×10^{-1}	4.01×10^{-1}	4.28×10^{-2}	.533	1.28×10^{0}	5.85×10^{-1} – 2.82×10^{0}	5.18×10^{-1} – 3.65×10^{0}
Alcohol	-9.62×10^{-3}	-9.23×10^{-3}	2.06×10^{-1}	-2.68×10^{-3}	.963	9.90×10^{-1}	6.62×10^{-1} – 1.48×10^{0}	6.18×10^{-1} – 1.62×10^{0}
FA	-9.21×10^{-3}	-9.53×10^{-3}	1.25×10^{-2}	-4.33×10^{-4}	.462	9.91×10^{-1}	9.69×10^{-1} – 1.02×10^{0}	9.62×10^{-1} – 1.02×10^{0}
ACC	**-5.76×10^{-1}**	**-6.43×10^{-1}**	**1.98×10^{-1}**	**-6.66×10^{-2}**	**.004**	**5.62×10^{-1}**	**3.81×10^{-1} – 8.28×10^{-1}**	**3.21×10^{-1} – 8.09×10^{-1}**
RelAge	**-6.96×10^{-2}**	**-7.89×10^{-2}**	**2.41×10^{-2}**	**-8.88×10^{-3}**	**.004**	**9.33×10^{-1}**	**8.90×10^{-1} – 9.80×10^{-1}**	**8.72×10^{-1} – 9.72×10^{-1}**
14	-3.36×10^{-1}	-3.72×10^{-1}	1.97×10^{-1}	-1.85×10^{1}	.089	7.15×10^{-1}	4.86×10^{-1} – 1.05×10^{0}	-7.48×10^{-1} – 1.43×10^{-1}
24	**-4.62×10^{-1}**	**-5.27×10^{-1}**	**2.12×10^{-1}**	**-5.25×10^{-1}**	**.029**	**6.29×10^{-1}**	**4.15×10^{-1} – 9.54×10^{-1}**	**-8.76×10^{-1} – 7.79×10^{-2}**

Note. Results of Cox proportional hazards regression analyses examining the predictive effect neural age defined with volume SBM components (model 1), neural age with covariates (model 2), and neural age with covariates and chronological age (model 3) on rearrest are presented. Model 1: Wald(2) = 11.87, p = .003, Likelihood Ratio(2) = 12.42, p = .002, Score(logrank)(2) = 12.02, p = .002. Model 2: Wald(9) = 23.56, p = .005; Likelihood Ratio(9) = 24.42, p = .004, Score(logrank)(9) = 24.96, p = .003. Model 3: Wald(10) = 29.3, p = .001; Likelihood Ratio(10) = 33.16, p = .001, Score(logrank)(10) = 32.97, p = .001. Bold variables are unique predictors within the model. RelAge is the participant's age when released from the correctional facility; PCL-R F1 and F2 refer to Factor 1 and Factor 2 scores from the Psychopathy Checklist – Revised (PCL-R) (Hare, 2003); PCL-R Int. refers to the PCL-R interaction term, formed by multiplying PCL-R Factor 1 and Factor 2 scores together; Drug refers to the participant's average use of the following drug classes: sedatives, cannabis, stimulants, opioids, cocaine, and hallucinogens collected from the Scheduled Clinical Interview for DSM-IV Axis I Disorders – Patient Version (SCID I/P) (M. B. First et al., 1997) and the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS) (Kaufman et al., 1997); Alcohol refers to the participant's average use of alcohol collected from the SCID I/P and K-SADS; FA refers to the false alarm rate to NoGo stimuli; ACC refers to dorsal anterior cingulate cortex mean activation. Component numbers are listed for SBM components included in the models.

Table 2c: Cox proportional hazards regressions with volume SBM components

Predictor	*B*	*Boot-strapped B*	*SE (B)*	*Boot-strapped SE (B)*	*p*-value	exp[B]	CI (95%) for exp[B]	*Boot-strapped CI (95%) for exp[B]*
Model 6 4	1.49×10^{-1}	1.59×10^{-1}	1.56×10^{-1}	1.71×10^{-1}	.338	1.16×10^{0}	8.55×10^{-1} – 1.58×10^{0}	-1.94×10^{-1} – 4.76×10^{-1}

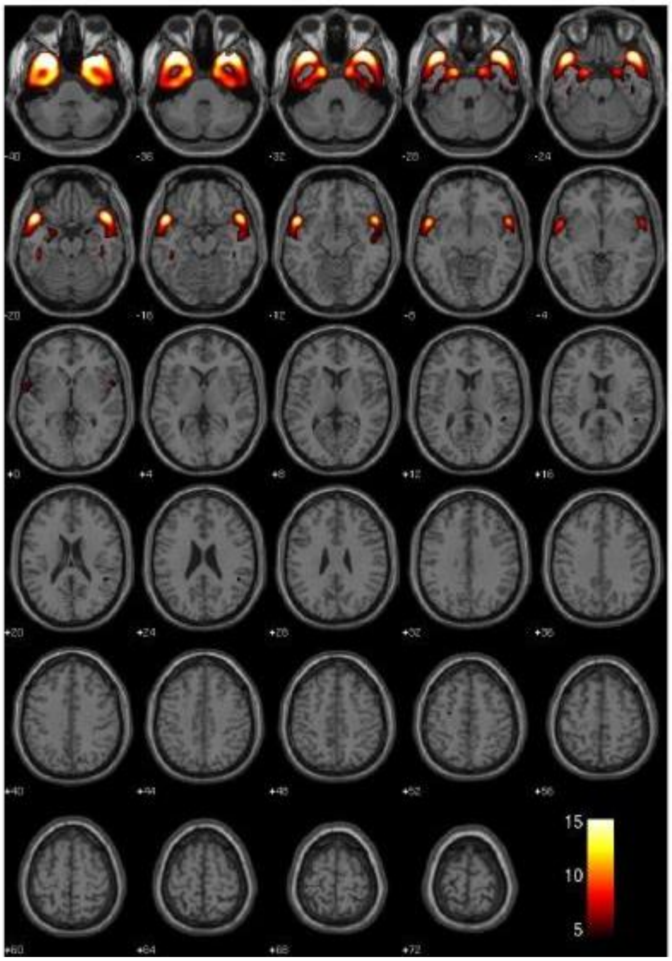
16	1.66×10^{-2}	1.53×10^{-2}	1.77×10^{-1}	1.92×10^{-1}	.925	1.02×10^{0}	7.19×10^{-1}– 1.44×10^{0}	-3.6×10^{-1} – 3.95×10^{-1}
21	**-4.03×10^{-1}**	**-4.21×10^{-1}**	**2.07×10^{-1}**	**2.21×10^{-1}**	**.052**	**6.69×10^{-1}**	**4.45×10^{-1} – 1.00×10^{0}**	**-8.18×10^{-1} – 5.00×10^{-2}**
24	2.15×10^{-1}	2.29×10^{-1}	1.53×10^{-1}	1.74×10^{-1}	.160	1.24×10^{0}	9.19×10^{-1} – 1.67×10^{0}	-1.4×10^{-1} – 5.41×10^{-1}
26	-8.2×10^{-2}	-8.74×10^{-2}	1.66×10^{-1}	1.66×10^{-1}	.621	9.21×10^{-1}	6.66×10^{-1} – 1.28×10^{0}	-4.02×10^{-1} – 2.49×10^{-1}
Model 7 PCL-R F1	-3.87×10^{-2}	-4.55×10^{-2}	7.04×10^{-2}	7.73×10^{-2}	.583	9.62×10^{-1}	8.38×10^{-1} – 1.10×10^{0}	-1.83×10^{-1} – 1.19×10^{-1}
PCL-R F2	**-2.57×10^{0}**	**-3.01×10^{0}**	**1.03×10^{0}**	**1.16×10^{0}**	**.012**	**7.69×10^{-2}**	**1.03×10^{-2} – 5.73×10^{-1}**	**-4.41×10^{0} – 1.40×10^{-1}**
PCL-R Int.	1.68×10^{-1}	1.88×10^{-1}	2.51×10^{-1}	2.83×10^{-1}	.503	1.18×10^{0}	7.24×10^{-1} – 1.93×10^{0}	-4.02×10^{-1} – 7.07×10^{-1}
Drug	1.85×10^{-1}	2.24×10^{-1}	4.55×10^{-1}	5.54×10^{-1}	.685	1.20×10^{0}	4.93×10^{-1} – 2.93×10^{0}	-9.42×10^{-1} – 1.23×10^{0}
Alcohol	5.75×10^{-2}	6.27×10^{-2}	2.16×10^{-1}	2.73×10^{-1}	.790	1.06×10^{0}	6.94×10^{-1} – 1.62×10^{0}	-4.79×10^{-1} – 5.91×10^{-1}
FA	-7.92×10^{-3}	-8.11×10^{-3}	1.42×10^{-2}	-7.96×10^{-5}	.578	9.92×10^{-1}	9.65×10^{-1} – 1.02×10^{0}	-3.65×10^{-2} – 2.16×10^{-2}
ACC	**-5.96×10^{-1}**	**-6.89×10^{-1}**	**1.95×10^{-1}**	**2.53×10^{-1}**	**.002**	**5.51×10^{-1}**	**3.76×10^{-1} – 8.08×10^{-1}**	**-9.99×10^{-1} – -8.2×10^{-2}**
4	1,27 x 10^{-1}	1.48×10^{-1}	1.86×10^{-1}	2.29×10^{-1}	.492	1.14×10^{0}	7.89×10^{-1} – 1.63×10^{0}	-3.41×10^{-1} – 5.59×10^{-1}
16	-4.53×10^{-2}	-4.69×10^{-2}	1.86×10^{-1}	2.47×10^{-1}	.807	3.42×10^{-10}	6.64×10^{-1} – 1.38×10^{0}	-5.29×10^{-1} – 4.39×10^{-1}
21	**-6.29×10^{-1}**	**-7.37×10^{-1}**	**2.71×10^{-1}**	**3.24×10^{-1}**	**.020**	**1.53×10^{-73}**	**3.13×10^{-1} – 9.06×10^{-1}**	**-1.17×10^{0} – 1.03×10^{-1}**
24	1.92×10^{-1}	2.21×10^{-1}	1.7×10^{-1}	2.18×10^{-1}	.260	1.24×10^{43}	8.68×10^{-1} – 1.69×10^{0}	-2.59×10^{-1} – 5.96×10^{-1}
Predictor	***B***	***Boot-strapped B***	***SE (B)***	***Boot-strapped SE (B)***	***p*-value**	**exp[B]**	**CI (95%) for exp[B]**	***Boot-strapped CI (95%) for exp[B]***
26	-5.44×10^{-2}	-6.18×10^{-2}	1.97×10^{-1}	2.11×10^{-1}	.782	1.10×10^{-13}	6.44×10^{-1} – 1.39×10^{0}	-4.59×10^{-1} – 3.69×10^{-1}
Model 8 PCL-R F1	-5.16×10^{-2}	-6.29×10^{-2}	7.63×10^{-2}	8.31×10^{-2}	.499	9.50×10^{-1}	8.18×10^{-1} – 1.10×10^{0}	7.95×10^{-1} – 1.10×10^{0}

PCL-R F2	**-2.79 x 10^{0}**	**-3.34 x 10^{0}**	**1.06 x 10^{0}**	**1.23 x 10^{0}**	**.008**	**6.16 x 10^{-2}**	**7.75 x 10^{-3} – 4.90 x 10^{-1}**	**2.69 x 10^{-3} – 3.52 x 10^{-1}**
PCL-R Int.	2.15 x 10^{-1}	2.43 x 10^{-1}	2.64 x 10^{-1}	3.03 x 10^{-1}	.415	1.24 x 10^{0}	7.39 x 10^{-1} – 2.08 x 10^{0}	7.09 x 10^{-1} – 2.38 x 10^{0}
Drug	2.99 x 10^{-1}	3.62 x 10^{-1}	4.69 x 10^{-1}	5.94 x 10^{-1}	.523	1.35 x 10^{0}	5.38 x 10^{-1} – 3.38 x 10^{0}	4.64 x 10^{-1} – 4.68 x 10^{0}
Alcohol	1.98 x 10^{-2}	1.95 x 10^{-2}	2.24 x 10^{-1}	2.84 x 10^{-1}	.930	1.02 x 10^{0}	6.58 x 10^{-1} – 1.58 x 10^{0}	5.95 x 10^{-1} – 1.77 x 10^{0}
FA	-1.23 x 10^{-2}	-1.36 x 10^{-2}	1.39 x 10^{-2}	1.56 x 10^{-2}	.377	9.88 x 10^{-1}	9.61 x 10^{-1} – 1.01 x 10^{0}	9.57 x 10^{-1} – 1.02 x 10^{0}
ACC	**-6.95 x 10^{-1}**	**-8.17 x 10^{-1}**	**2.03 x 10^{-1}**	**2.65 x 10^{-1}**	**.001**	**4.99 x 10^{-1}**	**3.36 x 10^{-1} – 7.43 x 10^{-1}**	**2.51 x 10^{-1} – 7.03 x 10^{-1}**
RelAge	**-5.47 x 10^{-2}**	**-6.49 x 10^{-2}**	**2.59 x 10^{-2}**	**3.12 x 10^{-2}**	**.035**	**9.47 x 10^{-1}**	**8.90 x 10^{-1} – 9.96 x 10^{-1}**	**8.78 x 10^{-1} – 9.92 x 10^{-1}**
4	1.53 x 10^{-1}	1.73 x 10^{-1}	1.82 x 10^{-1}	2.39 x 10^{-1}	.400	1.16 x 10^{0}	8.16 x 10^{-1} – 1.67 x 10^{0}	-1.06 x 10^{-1} – 1.67 x 10^{-2}
16	-1.32 x 10^{-1}	-1.49 x 10^{-1}	1.95 x 10^{-1}	2.65 x 10^{-1}	.496	8.76 x 10^{-1}	5.98 x 10^{-1} – 1.28 x 10^{0}	-3.33 x 10^{-1} – 6.01 x 10^{-1}
21	-7.81 x 10^{-1}	-9.2 x 10^{-1}	2.82 x 10^{-1}	3.46 x 10^{-1}	.005	4.58 x 10^{-1}	2.64 x 10^{-1} – 7.95 x 10^{-1}	-6.35 x 10^{-1} – 4.02 x 10^{-1}
24	1.56 x 10^{-2}	9.98 x 10^{-3}	1.97 x 10^{-1}	2.43 x 10^{-1}	.937	1.02 x 10^{0}	6.9 x 10^{-1} – 1.49 x $10^{\backslash 0}$	-1.32 x 10^{0} – 3.57 x 10^{-1}
26	-6.33 x 10^{-3}	-5.62 x 10^{-3}	1.93 x 10^{-1}	2.17 x 10^{-1}	.974	9.94 x 10^{-1}	6.81 x 10^{-1} – 1.45 x 10^{0}	-4.56 x 10^{-1} – 4.99 x 10^{-1}

Note. Results of Cox proportional hazards regression analyses examining the predictive effect neural age defined with volume SBM components (model 1), neural age with covariates (model 2), and neural age with covariates and chronological age (model 3) on rearrest are presented. Model 1: Wald(5) = 16.97, p = .005. Model 2: Wald(12) = 29.96, p = .003. Model 3: Wald(13) = 32.2 p = .001. Bold variables are unique predictors within the model. RelAge is the participant's age when released from the correctional facility; PCL-R F1 and F2 refer to Factor 1 and Factor 2 scores from the Psychopathy Checklist – Revised (PCL-R) (Hare, 2003); PCL-R Int. refers to the PCL-R interaction term, formed by multiplying PCL-R Factor 1 and Factor 2 scores together; Drug refers to the participant's average use of the following drug classes: sedatives, cannabis, stimulants, opioids, cocaine, and hallucinogens collected from the Scheduled Clinical Interview for DSM-IV Axis I Disorders – Patient Version (SCID I/P) (M. B. First et al., 1997) and the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS) (Kaufman et al., 1997); Alcohol refers to the participant's average use of alcohol collected from the SCID I/P and K-SADS; FA refers to the false alarm rate to NoGo stimuli; ACC refers to dorsal anterior cingulate cortex mean activation. Component numbers are listed for SBM components included in the models.

Figure 2 shows maps of the volume (IC 14 and 24) and density (IC 21) components that were significant predictors.

Figure 2a. Maps of the significant volume components (IC 14 (top) and 24 (bottom))



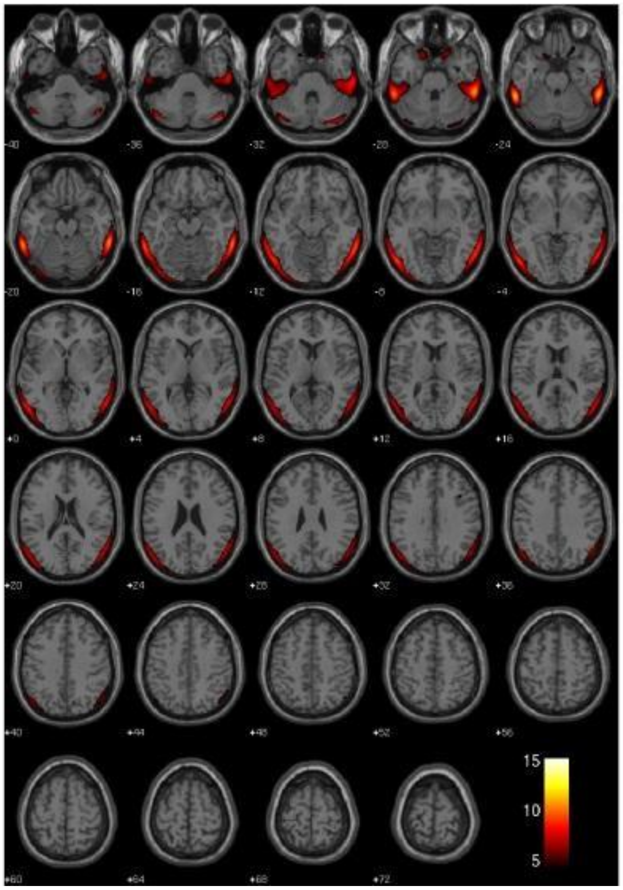
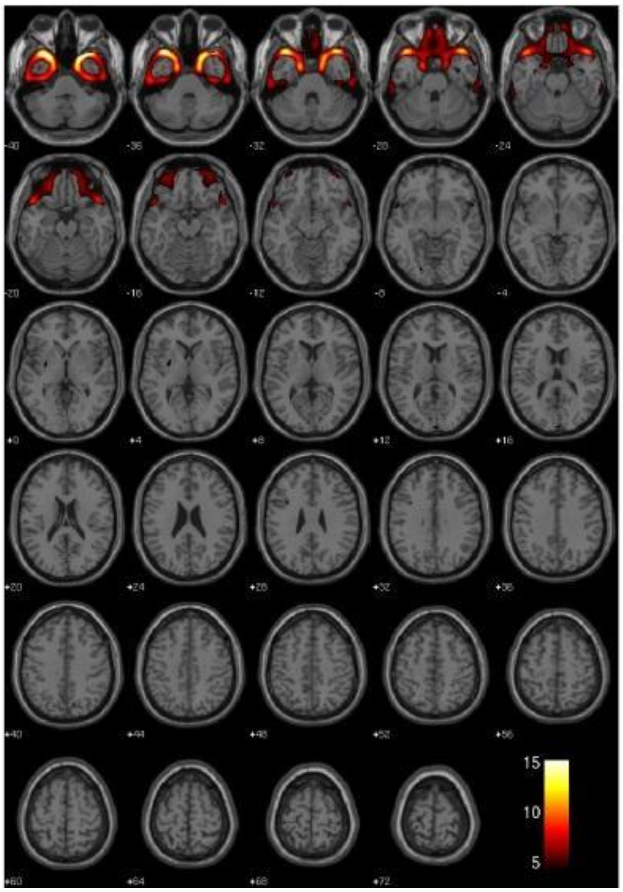

15
10
5

Figure 2b. *D*ensity (IC 21) component that predicted recidivism.



Using t-tests, less gray matter density was measured in the rearrested group, compared to the not-rearrested group, in Component 21 (temporal pole), $t(91) = 3.53$, $p < .001$. Components 12 (cerebellum) and 26 (occipital lobe) exhibited a similar, marginally significant, relationship, ($t(91) = 1.68$, $p = .097$, $t(91) = 1.85$, $p = .067$), respectively. More gray matter density was measured in the rearrested group, compared to the not-rearrested group, in component 4 (angular gyrus), 16 (inferior parietal gyrus) and 24 (cerebellum), $t(91) = -1.99$, $p = .049$, $t(91) = -2.16$, $p = .034$, $t(91) = -2.63$, $p = .010$, respectively. Component 20 (fusiform gyrus, calcarine fissure, lingual gyrus, & and occipital lobe) exhibited a similar, marginally significant, relationship, $t(91) = -1.77$, $p = .079$.

The pattern of AIC results is consistent with the known complexity of predicting recidivism. Simple models, that included only chronological age or brain age measures, had relatively high AIC values. Model 1 (chronological age only) was 405.62. Similarly, Model 3 (brain age volume) and Model 6 (brain age density) had high AIC's of 398.22 and 399.34, respectively. More complex models that included psychological variables and brain variables had better AIC values (Model 2 = 340.41; Model 4 = 337.19; Model 7 = 334.72; Model 5 = 330.44; Model 8 =332.27). In summary, Models (4, 5, 7 and 8) that included brain age measures and psychological variables were best supported. Likelihood ratio test and Score log rank test score are presented in the Tables. All reported p values indicate two-tailed tests.

Discussion

We confirmed our hypotheses that structural brain measures of age would distinguish offenders who are likely to re-offend from those who do not re-offend. Brain age (Models 3 and 6) accounted for more variance in the risk equation for rearrest than did chronological age (Model 1). Neural age incrementally added to risk outcomes (Models 5 and 8) and, when neural age is included in the model, chronological age is not necessary in predicting rearrest (Models 4 and 7). Reduced gray matter volume and density were identified as significant predictors of both neural age and rearrest. Specifically, reduced gray matter in bilateral anterior/lateral temporal lobes, amygdala, and inferior/orbital frontal cortex was helpful in predicting rearrest.

Temporal pole was identified in both volume and density measures as useful in predicting age and rearrest. Reduced gray matter volume and density in this region has been identified in psychopathic individuals (Cope et al., 2014; Ermer, Cope, Nyalakanti, Calhoun, & Kiehl, 2012;

Ermer, Cope, Nyalakanti, Calhoun, & Kiehl, 2013; Kiehl, 2006) and has been identified to be uniquely predictive of poor outcomes for such individuals (Cope et al., 2014). Immaturity of the amygdala, a structure well-known to be critical to affective processing, also predicted recidivism (Pardini, Raine, Erickson, & Loeber, 2014). Similarly, orbital frontal cortex is well understood to be a central component to regulating cognitive control. Taken together, these regions clearly highlight critical neural circuitry that contributes to problems in behavioral and affective control.

Importantly, a combination of variables was most useful in predicting rearrest. Psychopathy scores and anterior cingulate activity elicited from a Go/No Go task have demonstrated utility in predicting rearrest (Aharoni et al., 2013; Steele et al., 2015). By combining structural MRI data to these measures, several additional neural regions were identified to uniquely contribute to prediction models.

As with all studies, these findings carry a few limitations. First, our neural measure of age is uni-modal and could be improved upon. For example, there are many possible brain measures that can be used as a proxy for maturity/age (Brown et al., 2012; Cao et al., 2015; Dosenbach et al., 2010; Khundrakpam, Tohka, Evans, & Group, 2015; Mwangi, Hasan, & Soares, 2013). We recommend future work integrate multi-modal brain measures (structure, function, connectivity, diffusion etc.) to produce a more comprehensive "brain maturity index". Second, from a cost-benefit perspective, measuring chronological age is far simpler and less expensive than conducting brain imaging. What is demonstrated here is that neural measures accounting for age are as good or better than chronological age in our prediction models. This underscores a useful theoretical distinction between one's chronological age and one's brain maturity, which progresses at different rates in individuals for a variety of reasons. As chronological age ignores these differences, it is likely to miss some important variance in helping determine future outcomes. When the stakes are relatively low, these differences may not be practically important. However, when the stakes are high vis-à-vis predicting outcomes (i.e., civil commitment of dangerous sex offenders), it may be valuable to determine these differences with the utmost precision.

This is only an initial attempt of using neural age in models predicting behavioral outcomes. We believe future models, including those with multi-model measures and nonlinear terms, will account for more age-related variance and thus be more sensitive to specific outcomes. Moreover, future work could compute brain-psychopathy measures, brain-impulsivity measures, brain-IQ measures, brain-SES indicators, to aid in the neuroprediction of re-arrest. Indeed, it must be recognized that we included only a handful of known predictors of future re-arrest. It's possible that other measures, including genetics, other demographic or psychological data, may emerge as significant predictors in future studies.

In the coming years, the benefits of a more sensitive prediction model may necessitate an MRI scan, particularly when the interpretation of these models has high consequences. These findings are also valuable at the aggregative level for the development of interventions designed to modulate or train the endophenotypes that subserve behavioral inhibition and other protective skills. In this regard, this is the first paper to implicate the anterior and lateral temporal lobes and orbital frontal cortex in the prediction of future re-arrest (but see also (Pardini et al., 2014)) .

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Acknowledgments

This work was supported by the NIMH (R01 MH070539, R01 MH071896, R01 MH070539, PI Kiehl; R01 MH090169, PI: Kosson; R01 MH087525, PI: Decety), NIDA (R01 DA026964, R01 DA020870, R01 DA026505, PI: Kiehl), NIBIB (2R01EB000840, PI: Calhoun) and NICHD: 1R01HD082257-01 & 1R01HD092331-01 (PI: Kiehl) and funding from the MacArthur Foundation *Law & Neuroscience Project Phase I* (PIs Kiehl and Aharoni). These funding sources had no involvement in study design, data collection, writing of the manuscript, or decision to submit for publication.

We thank Elena Allen for her analytic support. We gratefully acknowledge the staff and inmates of the New Mexico Corrections Department and the Wisconsin Department of Corrections, for without their generous cooperation this work could not have been completed.

Author Contributions

K.A. Kiehl led the NIH/MacArthur projects that collected the data and conceived of the study approach. K.A., Kiehl, V. R. Steele, V. Rao, J.M. Maurer, V.D. Calhoun, E. Aharoni devised and implemented the analytical approach, performed data analysis, developed interpretations, and drafted the manuscript with contributions from all the other co-authors. M Koenigs coordinated data collection in Wisconsin prisons. All authors provided critical revisions and approved the final version for submission.

Competing financial interests

The authors do not declare any competing financial interests.